

ADVAIT ANIL KULKARNI

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Summary

Final-year Electronics & Telecommunication Engineering student with hands-on experience building microcontroller-based embedded and IoT systems on ESP32, Arduino. Proficient in Embedded C/C++, peripheral interfacing (UART, SPI, I2C, GPIO, PWM, ADC), sensor integration, and hardware-software debugging, demonstrated through multiple end-to-end embedded projects. Seeking an entry-level Embedded/Firmware Engineering role.

Education

KIT's College of Engineering (Empowered Autonomous), Kolhapur

Bachelor of Technology in Electronics and Telecommunication

Aug. 2023 – Expected 2027

CGPA: 8.3 (Final Year)

Technical Skills

Programming: Embedded C/C++, ARM Assembly

Embedded Platforms: ESP32, Arduino Uno, STM32, 8051, ARM7-LPC2148

Embedded/MCU Concepts: GPIO, Interrupts, Timers, PWM, ADC, Sensor Interfacing, Firmware Development

Communication Protocols: UART, SPI, I2C

Tools & Version Control: STM32CubeIDE, Keil uVision, Arduino IDE, Wokwi, Git/GitHub, VS Code

Applied Skills: Hardware-Software Integration, Hardware Debugging, Motor/Servo Control

Projects

Mecanum Wheel Based Robotic Arm | *Arduino IDE, MIT App Inventor, Arduino Uno*

October 2024

- Engineered Embedded C firmware on Arduino Uno to coordinate 4 servo motors for robotic arm motion control.
- Designed a mecanum-wheel omnidirectional base with integrated servo/motor control logic.
- Interfaced an HC-05 Bluetooth module with the microcontroller for wireless control via a companion Android app.
- Performed hardware debugging and PWM-based servo calibration to improve motion accuracy and stability.
- 🐙 github.com/Advait-Kulkarni/Mecanum-Wheel-Based-Robotic-Arm

IoT-Based Incident Detection System | *ESP32, Embedded C++, MPU6050, GPS, GSM, UART, I2C*

May 2026

- Designed an ESP32-based wearable incident-detection system for mountain climbers and elderly users.
- Implemented MPU6050-based fall/impact detection via I2C interfacing and threshold-based firmware logic.
- Integrated GPS and GSM (SIM800L) modules over UART for location tracking and emergency SMS alerts.
- Validated sensor, GPS, and GSM communication through structured hardware debugging and testing.
- 🐙 github.com/Advait-Kulkarni/Incident-Detection-System

Smart Parking System | *ESP32, Arduino IDE, Embedded C++, PHP, MySQL*

May 2025

- Developed ESP32 firmware for automated gate control using real-time ultrasonic sensor data.
- Integrated ESP32 controller with a PHP/MySQL backend for QR-based entry/exit verification.
- Designed firmware-level fault handling for invalid entries and communication failures.
- Debugged sensor interfacing and communication logic for reliable real-time operation.
- 🐙 github.com/Advait-Kulkarni/Smart-Parking-System

Certifications

- ARM Cortex (STM32) Fundamentals: Building Embedded Systems [Link](#)
- Embedded C Programming Essentials [Link](#)
- Embedded Systems Using C [Link](#)
- Problem Solving through Programming in C [Link](#)

Co-Curricular Activities

Core Member, Team Mavericks

August 2024 – Present

- Organized **Bodhantra**, a 5-day professional development event for first-year students, and played a role of event lead in organizational activities for **INVICTA**, a flagship technical event of the college.
- Coordinated **Verbafest** and participated in school outreach visits to promote technical awareness among students.

Honors And Awards

- Secured **First Place** in **three national-level technical project competitions**. [Link](#)
- First Runner up - Pitching Deck (National Idea Pitching Competition), KIT College of Engineering, Kolhapur [Link](#)